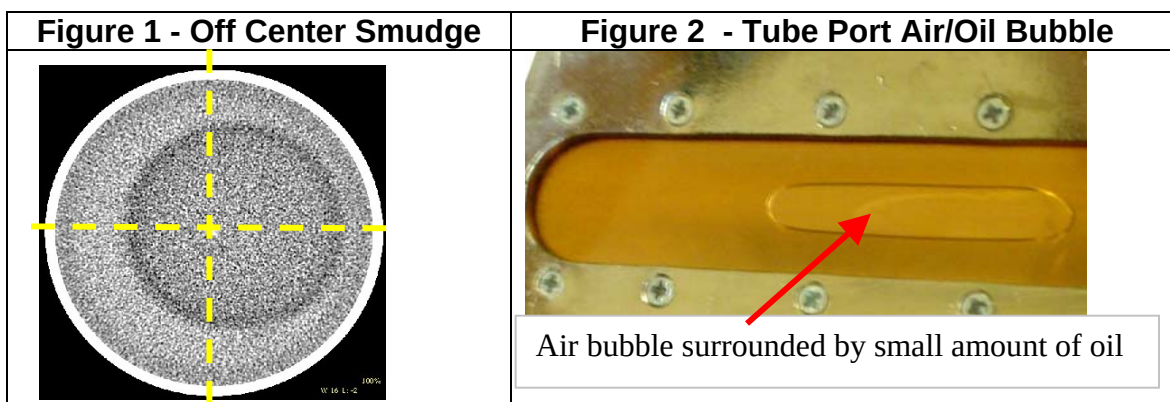


FCT SERVICE NOTE SN06102121a

Class C Service information enclosed.

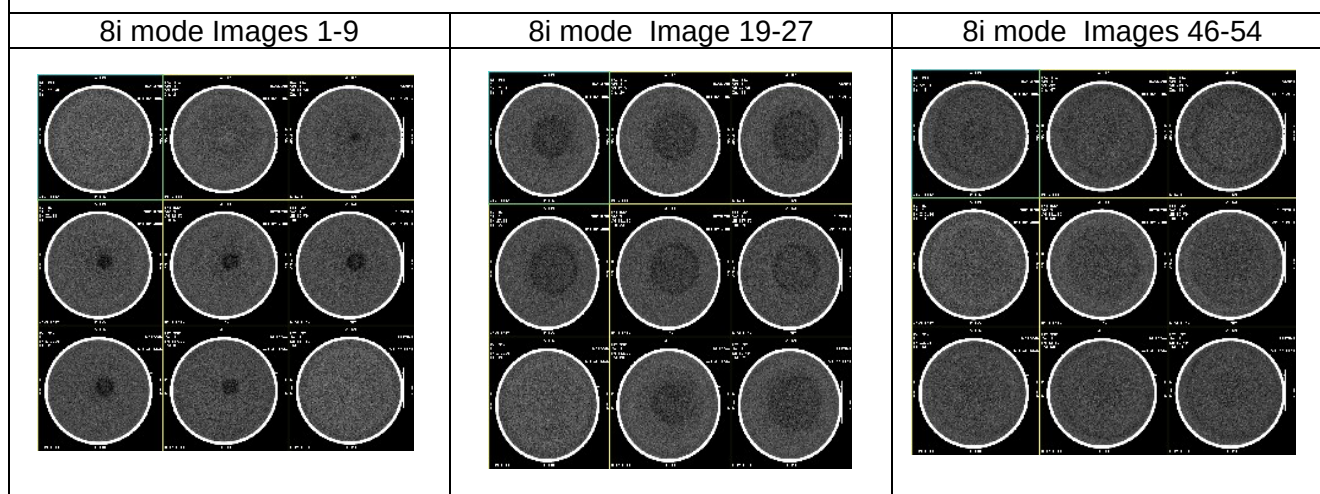
Date: Oct.25, 2006
Effectivity: LightSpeed 5.x (HP80, HP100), 7.x (VCT) and RT¹⁶
Subject: "Off Center Smudge" - Image Artifact



Details: Off Center Smudge - Image Artifact

Symptoms: Please take a moment and view the image artifact above. The artifact is typically to the right of ISO center. Notice that the artifact is round in appearance and is off center. The artifact is intermittent in nature and tends to change in size and CT number from one rotation to the next. The artifact may disappear as the Gantry continues to rotate.

Figure 3: Image 1 – 54 Artifact tends to disappear as the Gantry continues to rotate.



Cause: The cause for the artifact is oil which has leaked into the tube port window and changes the filtration of the x-ray beam. After about one minute of Gantry rotation, the bubble settles and the artifact tends to disappear. Therefore, it usually appears only in the first few image series. Sometimes the oil in the Tube Port can be seen on the top of the copper filter, looking through the collimator (as seen in figure 2).

GE Healthcare Class C eService Notes are copyright (c) to General Electric Company Inc with all rights reserved. Class C eService Notes issued by GE Healthcare are proprietary to GE Healthcare Systems. Disclosure to persons not employed by GE Healthcare or not in possession of a Advanced Service Agreement with GE Healthcare is prohibited.

Duplication and distribution of any class of eService Notes, in any form by entities other than GE Healthcare is strictly prohibited.

QA Phantom Check: If such an artifact is suspected, wait 15 minutes with the tube parked at the normal 2 O'clock position, take a series of 1.0 second axial scans, see details below, take note VCT and HP-100 scan Techniques are slightly different, mainly **4i vs. 8i** (120kv 200mA, 200 images). Review the images for the artifact as seen in figure 3.

- 1) Select > New Patient > Service Generic Scan 42.1
- 2) Place the 20 cm Phantom with the water section in the beam – center with Alignment lights
- 3) Select appropriate for scan technique for your scanner (VCT or H-Power)

- VCT - 8i x 5mm (40mm) 120 200ma 1sec Axial with an ISD of 4 seconds set the interval to zero
- H-Power - 4i x 5mm (20mm) 120 200ma 1sec Axial with an ISD of 4 seconds set the interval to zero

- 4) Continue to ADD Group via the Add Group button until 200 images are selected

VCT 8I x 5mm (40mm) I-17.50 S-17.5 Interval= 0 ISD set to 4 sec Service Generic 42.1

Images	Scan Type	Start Location	End Location	No. of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	Prep Group (sec)	ISD (sec)	Breath Hold (sec)	Breathe Time (sec)	Voice Lights Timer	Cine Duration (sec)
1-8	Axial Full 1.0 sec	I17.500	S17.500	8	5.0 8i	0.000	S0.0	Head	120	200	1.0	0.0	4.0	N	N	N	2.0
9-16	Axial Full 1.0 sec.	I17.500	S17.500	8	5.0 8i	0.000	S0.0	Head	120	200	1.0	1.0	4.0	N	N	N	2.0
17-24	Axial Full 1.0 sec.	I17.500	S17.500	8	5.0 8i	0.000	S0.0	Head	120	200	1.0	1.0	4.0	N	N	N	2.0

H-Power 4I x 5mm (20mm) I-7.50 S-7.50 Interval= 0 ISD set to 4 sec Service Generic 42.1

Images	Scan Type	Start Location	End Location	No. of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	Prep Group (sec)	ISD (sec)	Breath Hold (sec)	Breathe Time (sec)	Voice Lights Timer	Cine Duration (sec)
1-4	Axial Full 1.0 sec	I7.500	S7.500	4	5.0 4i	0.000	S0.0	Head	120	200	1.0	0.0	4.0	N	N	N	2.0
5-8	Axial Full 1.0 sec.	I7.500	S7.500	4	5.0 4i	0.000	S0.0	Head	120	200	1.0	1.0	4.0	N	N	N	2.0
9-12	Axial Full 1.0 sec.	I7.500	S7.500	4	5.0 4i	0.000	S0.0	Head	120	200	1.0	1.0	4.0	N	N	N	2.0

GE Healthcare Class C eService Notes are copyright (c) to General Electric Company Inc with all rights reserved. Class C eService Notes issued by GE Healthcare are proprietary to GE Healthcare Systems. Disclosure to persons not employed by GE Healthcare or not in possession of a Advanced Service Agreement with GE Healthcare is prohibited.

Duplication and distribution of any class of eService Notes, in any form by entities other than GE Healthcare is strictly prohibited.

Figure 4: Window Screw FMI Complete Sticker

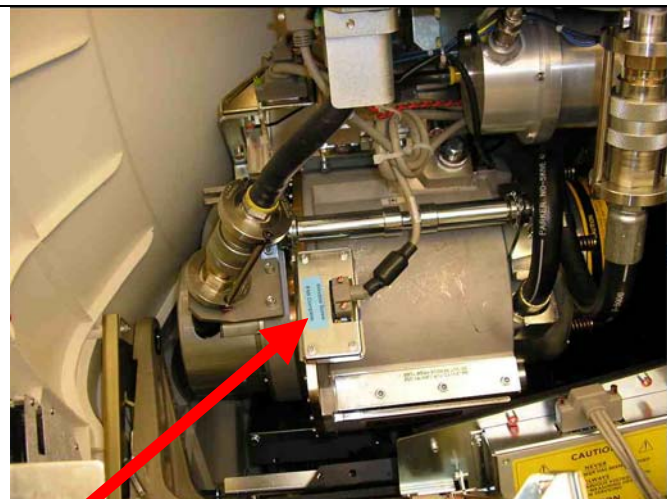


Figure 5: Window Screw FMI Complete Sticker -Close-up



Resolution: CT and Tubes Engineering is currently validating a proposed FMI. If you see the artifact please call the numbers below for the latest information.

The most recently manufactured tubes include a blue sticker as depicted above indicating the FMI is already complete on that unit.

Questions:

Call the OLC OR
John Breunissen EA
Ryan August
David Palermo OLC
Tom Hurdis EA

cell (414) 202 9485 or 414 647 4130
cell (414) 745 8127 or (414) 647 4281
cell (262) 853 8438 or (262) 524-5166
cell 262 510-5782

David Palermo
OLC Services